



**UNIVERSITI KUALA LUMPUR KAMPUS KOTA MALAYSIAN
INSTITUTE OF INFORMATION TECHNOLOGY**

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Lecturer	ASSOC. PROF. DR. MEGAT NORULAZMI
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Student Name & ID	NUR HUMAIRA BINTI ABDUL WAHAB (52224123318) NUR ZAYANEE MAHFUZAH BINTI ZAYZAMRY (52224123241)
Assessment	GROUP PROJECT ASSIGNMENT REPORT

YouTube Presentation Link:

(LINK SINI)

1. Introduction

The rapid growth of streaming platforms has transformed the entertainment industry by providing users with instant access to a wide range of digital content. Netflix is one of the world's leading streaming services, offering thousands of movies and television shows across different genres, countries, and audience categories.

This project aims to perform exploratory data analysis (EDA) on the Netflix Titles dataset obtained from Kaggle. Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn within Jupyter Notebook, the dataset was cleaned, analyzed, and visualized to identify patterns, trends, and insights related to content distribution, production countries, ratings, release years, and genres. The findings from this analysis can help support data-driven decision-making regarding content strategy and audience engagement.

2. Objectives

- To analyze the distribution of movies and TV shows available on Netflix.
- To identify the major countries contributing content to the Netflix platform.

- To examine content ratings and target audience categories.
- To investigate content release trends and Netflix content growth over time.
- To analyze the most popular genres available on Netflix.
- To generate actionable insights based on the findings obtained from the dataset.

3. Import Libraries

The following Python libraries were imported to support data processing, visualization, and analysis:

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

- Pandas was used for data manipulation and analysis.
- NumPy was used for numerical operations.
- Matplotlib was used for data visualization.
- Seaborn was used to create enhanced statistical graphics.

4. Load Dataset

The Netflix dataset was loaded into Jupyter Notebook using Pandas.

```
[2]: df = pd.read_csv("netflix_titles.csv")
```

```
[3]: df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...

The dataset contains information about Netflix content, including title, type, director, cast, country, release year, rating, duration, genre, and description.

5. Dataset Overview

```
[4]: df.shape
```

```
[4]: (8807, 12)
```

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   str
1   type            8807 non-null   str
2   title           8807 non-null   str
3   director        6173 non-null   str
4   cast            7982 non-null   str
5   country         7976 non-null   str
6   date_added      8797 non-null   str
7   release_year    8807 non-null   int64
8   rating          8803 non-null   str
9   duration        8804 non-null   str
10  listed_in       8807 non-null   str
11  description      8807 non-null   str
dtypes: int64(1), str(11)
memory usage: 825.8 KB
```

```
[6]: df.columns
```

```
[6]: Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
        'release_year', 'rating', 'duration', 'listed_in', 'description'],
        dtype='str')
```

The dataset contains 8,807 records and 12 attributes. Most columns contain textual data, while the release year column is stored as an integer.

The dataset attributes include:

- show_id
- type
- title
- director
- cast
- country
- date_added
- release_year
- rating
- duration
- listed_in
- description

6. Data Cleaning

6.1 Missing Values

Missing values were identified using the following command:

```
[7]: df.isnull().sum()
```

```
[7]: show_id      0
     type       0
     title      0
     director   2634
     cast       825
     country    831
     date_added  10
     release_year 0
     rating     4
     duration   3
     listed_in  0
     description 0
     dtype: int64
```

Several columns contained missing values, with the director column having the highest number of missing entries.

To simplify the analysis and prevent errors during processing, all missing values were replaced with the label "Unknown".

```
[8]: df.fillna("Unknown", inplace=True)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalané, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...
...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	July 1, 2019	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...

8807 rows x 12 columns

A second validation confirmed that no missing values remained in the dataset.

```
[9]: df.isnull().sum()
```

```
show_id      0
type         0
title        0
director     0
cast         0
country      0
date_added   0
release_year 0
rating       0
duration     0
listed_in    0
description   0
dtype: int64
```

6.2 Duplicate Records

Duplicate records were checked using:

```
[10]: df.duplicated().sum()  
[10]: np.int64(0)
```

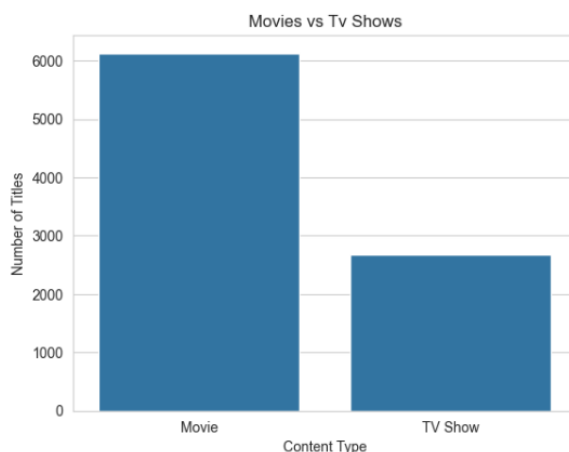
No duplicate records were found in the dataset. Therefore, no duplicate removal process was required.

7. Exploratory Data Analysis (EDA)

Several visualizations were generated to understand content patterns and trends within the Netflix catalog.

7.1 Distribution of Content Type

```
[11]: sns.countplot(data=df, x='type')  
plt.title('Movies vs Tv Shows')  
plt.xlabel('Content Type')  
plt.ylabel('Number of Titles')  
plt.show()
```



Clear Dominance: The storyline highlights a major difference in the platform's collection, demonstrating that films are far more common than TV series.

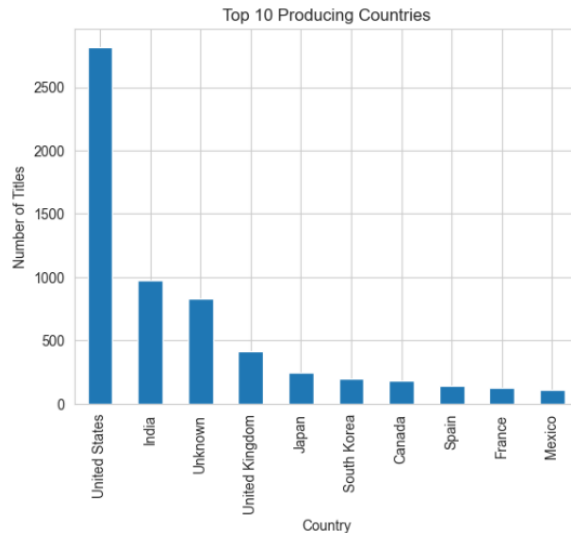
Large Margin: The bar for movies is more than two times as tall as the bar for TV series, highlighting the platform's strong preference for movie over TV show.

Library Focus: This distribution shows that building a large film archive has traditionally been prioritized over acquiring current television content.

7.2 Top Content Producing Countries

```
[28]: df['country'].value_counts().head(10).plot(kind='bar')

plt.title('Top 10 Producing Countries')
plt.xlabel('Country')
plt.ylabel('Number of Titles')
plt.show()
```



Overwhelming Leadership: The United States is the clear leader in content production on the platform. Its volume is much higher than all other countries in the dataset.

Secondary Tier: India ranks second and has a clear lead over the remaining countries.

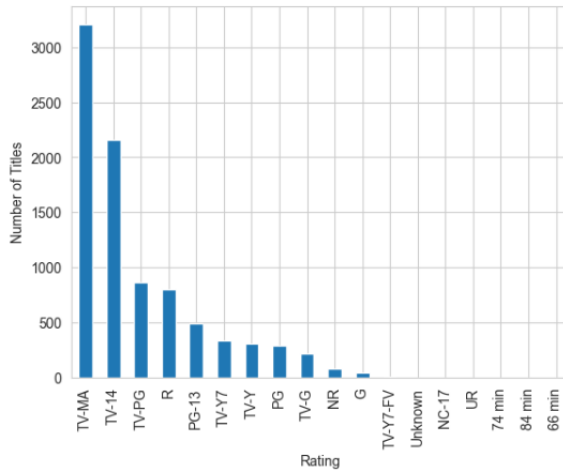
Data Limitation: A large portion of the content is labelled as “Unknown.” This category ranks close to India, which suggests that a significant amount of geographic information is missing or not recorded.

Gradual Decline: After the United Kingdom, the other top content-producing countries, mainly from Asia such as Japan and South Korea, North America such as Canada and Mexico, and Europe such as Spain and France, show a steady decrease in content volume, forming a long tail of smaller contributors.

7.3 Content Ratings Distribution

```
[29]: df['rating'].value_counts().plot(kind='bar')

plt.xlabel('Rating')
plt.ylabel('Number of Titles')
plt.show()
```



Dominance of Mature Content: The plot shows that TV-MA is the most frequent rating on the platform. This indicates a strong concentration of content intended for mature adult audiences.

Strong Teen Presence: TV-14 is the second most common rating and has a clear lead over most other categories. This suggests that a large portion of the content is targeted at teenagers.

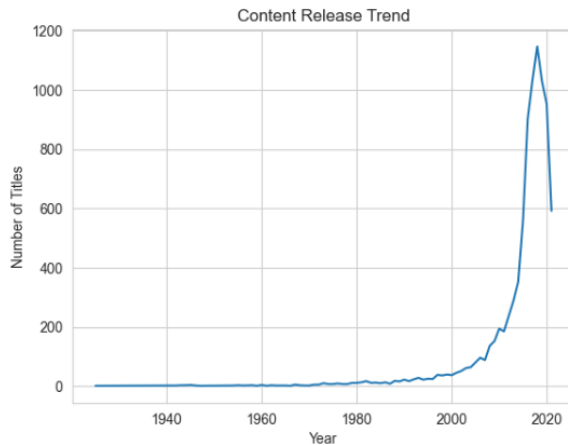
Moderate Mid-Tier Ratings: Ratings such as TV-PG, R, and PG-13 form a middle group with moderate and similar representation. These categories mainly include content for older children, families, and general audiences.

Minority and Anomalous Data: Content aimed specifically at young children, such as TV-Y, TV-Y7, and G, makes up a relatively small portion of the library. At the lower end of the chart, there are also a few inconsistencies where values like “74 min” or “84 min” appear, suggesting formatting issues where runtimes were mistakenly classified as ratings.

7.4 Release Year Trend

```
[30]: df['release_year'].value_counts().sort_index().plot()

plt.title('Content Release Trend')
plt.xlabel('Year')
plt.ylabel('Number of Titles')
plt.show()
```



Long Period of Inactivity: For several decades in the mid to late 20th century, content volume remained low and relatively flat. This suggests that only a small number of older or historical titles were available.

Gradual Growth: Around the turn of the century, there is a slight upward trend. This indicates a slow but steady expansion of the library as more modern content began to be added.

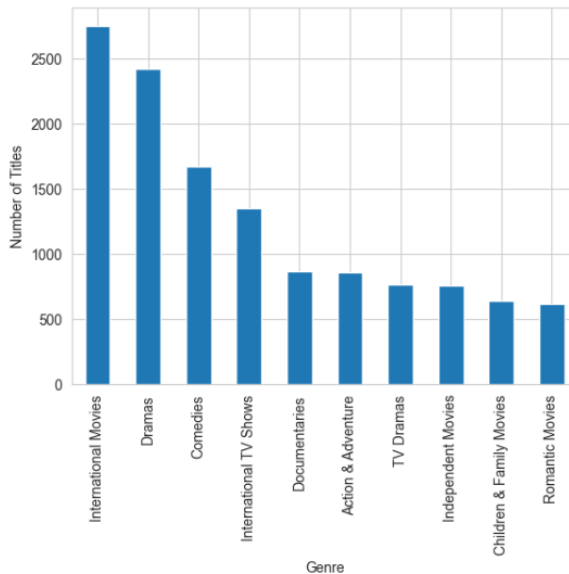
Exponential Surge: In the most recent decade, the chart shows a very sharp and dramatic increase. This reflects a period of rapid growth where most of the platform's content was released.

Recent Drop-off: At the very end of the graph, there is a sudden decline. This is likely due to a data issue, such as incomplete records for the final year or inconsistencies in data collection during that period.

7.5 Genre Analysis

```
[15]: genres = df['listed_in'].str.split(', ').explode()
genres.value_counts().head(10).plot(kind='bar')

plt.xlabel('Genre')
plt.ylabel('Number of Titles')
plt.show()
```



Leading Categories: International Movies and Dramas emerge as the most dominant genres on the platform by a substantial margin, capturing the largest share of the overall content catalog.

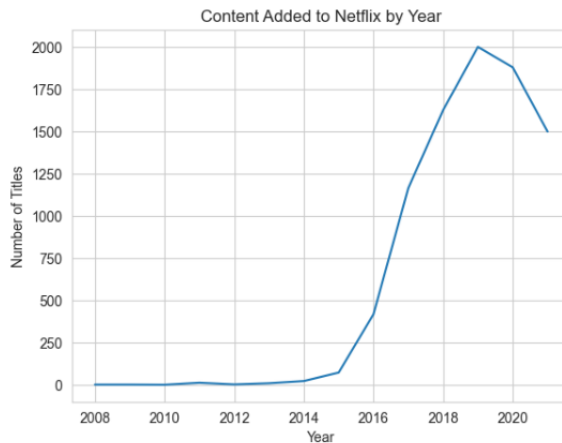
Strong Mid-Tier Content: Comedies and International TV Shows form a solid secondary tier of popular offerings, comfortably outpacing the lower-ranked genres.

Global vs. Domestic Focus: The high placement of both International Movies and International TV Shows underscores the platform's heavy reliance on non-domestic or globally sourced content to flesh out its library.

Evenly Distributed Tail: The remaining genres including Documentaries, Action & Adventure, TV Dramas, Independent Movies, Children & Family Movies, and Romantic Movies show a very gradual, uniform decline, representing a stable and balanced variety of niche or specific content categories at the lower end of the top ten.

7.6 Content Added to Netflix by Year

```
[16]: df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
      df['added_year'] = df['date_added'].dt.year
      df['added_year'].value_counts().sort_index().plot()
      plt.title('Content Added to Netflix by Year')
      plt.xlabel('Year')
      plt.ylabel('Number of Titles')
      plt.show()
```



Flat Early Baseline: For the first several years shown on the graph, the volume of newly added titles remained completely flat and close to zero, representing a period when content acquisition on the platform was minimal.

Initial Growth Spark: A slight upward trend begins to take shape in the mid-2010s, marking the initial phase of the platform's expansion into building a larger digital library.

Rapid Acceleration: Following this initial spark, the line transitions into a remarkably steep, rapid climb. This indicates an aggressive, multi-year content acquisition and production boom where massive volumes of content were uploaded annually.

Peak and Recent Decline: After reaching its highest point, the graph takes a distinct downward turn in the final years. This recent drop-off could point to a strategic shift toward quality over quantity, global content production slowdowns, or an incomplete data tracking window for the final calendar year.

8. Key Findings

8.1 Content Type Imbalance

The platform's library is heavily dominated by movies, with the number of films more than double the number of TV shows. This indicates a strong preference for standalone content over episodic series.

8.2 Geographic Centralization

Content production is largely concentrated in the United States, with India as the second major contributor. However, a significant portion of the dataset is labeled as “Unknown,” indicating incomplete geographic metadata.

8.3 Target Audience Focus

The platform primarily targets mature and teenage audiences, with TV-MA and TV-14 being the most frequent ratings. Content designed for young children represents only a small portion of the overall catalog.

8.4 Modern Content Dominance

The dataset is largely composed of modern titles, with a sharp increase in content released during the 2010s. Older 20th-century content remains limited, showing a strong focus on recent productions.

8.5 Growth Trend Over Time

Content additions increased rapidly during the 2010s, reaching a peak before declining in recent years. This reflects a shift from aggressive expansion to slower growth.

8.6 Strong International Presence

International content forms a key part of the catalog, with categories such as “International Movies” and “Dramas” ranking highly in volume.

9. Actionable Insights

9.1 Improve Content Balance

Increase investment in TV series to improve long-term user engagement and retention, as episodic content typically drives sustained viewing compared to standalone films.

9.2 Diversify Content Sources

Reduce reliance on the United States by expanding content production and licensing in emerging international markets, leveraging the strong performance of international categories.

9.3 Strengthen Data Quality Controls

Implement stricter data validation during ingestion to prevent errors such as runtime values (e.g., “74 min”) being incorrectly stored in rating fields.

9.4 Enhance Metadata Completeness

Enforce mandatory geographic tagging for all content entries to reduce the number of “Unknown” classifications and improve analytical accuracy.

9.5 Optimize Content Strategy

Shift focus on high-quality original productions, particularly in mature and teen genres, in response to the recent slowdown in overall content additions.

10. Conclusion

The results revealed that Netflix's catalog is dominated by movies, heavily influenced by content from the United States, and primarily targeted toward mature and teenage audiences. Additionally, the platform experienced substantial content growth throughout the 2010s before showing signs of slower expansion in recent years.

Based on the findings, several recommendations were proposed, including increasing investment in TV series, expanding international content production, improving metadata quality, and strengthening data governance practices. These actions can help improve catalog diversity, enhance analytical reliability, and support long-term strategic growth.